PATRICK SLIZ

psliz@college.harvard.edu | linkedin.com/in/patrick-sliz | patricksliz.com

EDUCATION

Harvard College, Cambridge, MA — GPA: 3.8/4.0 **May 2027**
A.B. Candidate in Computational Neuroscience. Secondary in Economics.
- Relevant Coursework: Computational Neuroscience, Biological Signal Processing, Memory Models, Linear Algebra

St. Sebastian's High School, Needham, MA — Valedictorian, 1560 SAT. **May 2023**

RESEARCH EXPERIENCE

Nia Therapeutics, Boston, MA **October 2026 - Present**
Research Intern

Harvard Medical School / Massachusetts General Hospital, Boston, MA **January 2026 - May 2026**
Undergraduate Researcher, Ziv Williams Laboratory
- Decoded speech and language from intracranial single-neuron (Neuropixels) recordings in patients undergoing awake neurosurgical monitoring; built the analysis pipelines in Python
- Fit ridge encoding models mapping all 13 layers of GPT-2 onto prefrontal single-neuron firing rates across three patient cohorts, with permutation-based significance testing (2,000 shuffles)
- Diagnosed a generalization failure in the decoding pipeline as a word- vs. sentence-level representational mismatch; rebuilt the train/test framework to correct it

Boston Children's Hospital, Manton Center for Orphan Disease Research, Boston, MA **December 2025 - January 2026**
Winter Research Intern
- Used Google's AI Co-Scientist to generate and evaluate protein-protein interaction hypotheses from biological datasets

University of Warsaw, **Department of Quantitative Finance,** Warsaw, Poland **Summers 2024 - 2025**
Research Internship with Professor Piotr Wójcik
- Built ML models (LASSO, SVR, random forest, XGBoost, LightGBM) in R and Python predicting state and county GDP and income from night-time satellite imagery; interpreted feature importance with SHAP and permutation methods
- Co-authored the resulting paper, "Where the Light Is," submitted to the Journal of Regional Science (2026)

Stanford University, Neurosciences Institute, HHMI Laboratory, Palo Alto, CA **Summers 2022 - 2023**
Software Developer in Cryo-electron Tomography (Cryo-ET) with Professor Axel Brunger
- Developed a subtomogram-averaging pipeline for high-resolution molecular imaging; rebuilt the lab's MATLAB toolkit as PyCrESTA, a Python/Kivy application, and optimized 3D signal-processing routines with multi-threading and parallelization
- Built visualization, parsing, and cross-correlation tools used by lab members in collaboration with structural biologists

PROFESSIONAL EXPERIENCE

Cornerstone Research, *Incoming Analyst*, Boston, MA **Starting 2027**

GoTo, Boston, MA **June 2026 - August 2026**
Strategic Finance Intern, Business Planning & Analysis Team (part-time through December 2026)
- Built an ARR forecasting suite of six time-series and ML models (Prophet, ETS, MSTL, DHR-ARIMA, LightGBM, XGBoost) covering ~$100M in ARR; cut backtested forecast error (MAPE) from ~50% to ~14%; deployed to Databricks
- Shipped two production AI agents: an accounting agent reconciling sales tax data across four sources, replacing manual reconciliation, and a QBR agent generating deck cheat sheets and post-meeting analyses with human-in-the-loop gating
- Co-authored an AI strategy white paper with the VP of Strategic Finance on adoption priorities and AI pricing economics across GoTo's product portfolio; incorporated feedback from Product and Strategy stakeholders; reviewed by CFO and CCO

LEADERSHIP EXPERIENCE

The Harvard Independent, *Vice President* **January 2024 - Present**
- Co-lead operations and finances for a 140+ person organization alongside the President; own merchandise and program revenue
- Grew website readership >350% to 444k+ views in a single semester via SEO and infrastructure overhaul; built a Python pipeline automating weekly issue publication from Google Drive to WordPress

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), R, MATLAB, C, Bash, Git
Methods: Time-series forecasting, Ridge/LASSO/logistic regression, PCA, SVM, LDA, permutation testing, SHAP, ASR
Finance & Analytics: Excel, SQL, Power BI, Databricks, Stata, NetSuite
Languages: English (native), Polish (fluent), Spanish (conversational)